

The Digital Twins of Thor's Hammer Based on Motion Sensing

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ABSTRACT

Ancient humans attribute the phenomenon of thunder and lightning to divine power. The power of Thor that can lift Thor's Hammer, the body not be hurt by thunder and lightning. It's not impossible for us to control thunder and lightning like Thor. The Digital Twins system of the robotic arm designed in this paper integrates the physical device of the robotic arm, the digital model of robotic arm, the body sense interaction, and the virtual-reality mapping module. It can digitally control the robotic arm. With this system, we can all lift Thor's hammer in the future.

Keywords: Digital Twins, wireless power transmission, virtual reality, robotic arm, motion sensing interaction.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interactive systems and tools—User interface management systems

1 INTRODUCTION

Thor can control thunder and lightning, and it is not impossible for us to achieve it. This system takes this as a starting point and based on Digital Twins technology to design the Robotic arm body sense control system. The system lift Thor's Hammer by controlling the robotic arm through motion sensing to guide the thunder and lightning to the designated position. It is worth mentioning that the system chooses Tesla coil to create lightning. Although the current power supply service is generally wired, the traditional cable way has some disadvantages and limitations, such as equipment aging, electricity safety, environmental restrictions and so on. Wireless power transmission uses the ionosphere as a conductor, and transmits electrical energy through electromagnetic wave oscillations, which can transmit electrical energy to all parts of the world with extremely low loss. We believe that, wireless transmission will completely replace wired transmission in the future, whether in the real world or the Metaverse. Therefore, this system uses Dual Resonant Solid State Tesla Coil(DRSSTC) to transmit wireless power to create lightning effects and simulate wireless power transmission.

2 DESIGN OF DIGITAL TWINS SYSTEM OF ROBOTIC ARM

The system first performs the modeling of the robotic arm and constructs its digital model in the virtual space. PID is used to control the robotic arm, and ROS is used as the interaction of network data. The communication system uses the CAN architecture to transmit data and instructions to realize interaction of virtual and real world.

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2.1 Robotic Arm Model Establishment

Unified Robot Descriptin Format(URDF) is a robotic arm model file described in XML format [1], similar to D-H parameters, which includes parameters Links and Joints. This system models the six-degree-of-freedom manipulator used. In Solidworks, 1:1 to the actual manipulator. The specific steps are shown in Figure 3.

2.2 Robotic Arm Control

For the control of the robotic arm, we adopted the enhanced particle swarm optimization (PSO) method for the optimization of fractional controllers. This method was proposed by M. Zamani et al. [2] uses time domain and frequency domain specifications and proposed a new cost function. The definition of performance indicators is as follows:

$$J(k) = w_1 M_p + w_2 t_e + w_3 t_{ss} + w_4 E_{ss} + \int_0^1 (w_5 |e(t)|) + w_6 u^2(t) dt + \frac{w_7}{PM} + \frac{w_8}{GM} \quad (1)$$

After calculation, we found that M_p , PM , GM in the above formula have little effect on the function, so we introduce a constant to replace these three parameters. The new cost function is as follows:

$$J(k) = w_1 t_e + w_2 t_{ss} + w_3 E_{ss} + \int_0^1 (w_4 |e(t)|) + w_5 u^2(t) dt + \varphi \quad (2)$$

We used the functions before and after the improvement to test the response time of different steering gears of the robotic arm. The test results are as follows:

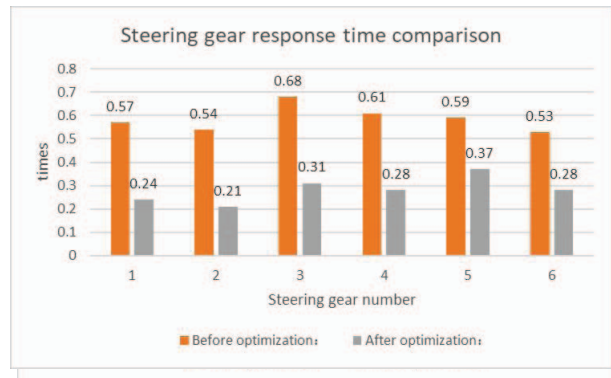


Figure 1: Comparison of response time before and after algorithm optimization

2.3 Robotic Arm Motion Sensing Control

In order to increase the user's immersive experience, we have added Kinect-based motion sensing control to this system. The system uses CMOS infrared sensors to perceive the surrounding environment in a black and white spectrum. Use Light Coding technology to mark the space, judge the direction and amplitude of the movement according to the image change of the target object, and recognise the "Gesture" of Thor's Hammer. And through coordinate calculation, it is transformed into Cartesian coordinates required by the robotic arm. The robotic arm makes the corresponding action after receiving the coordinate signal.

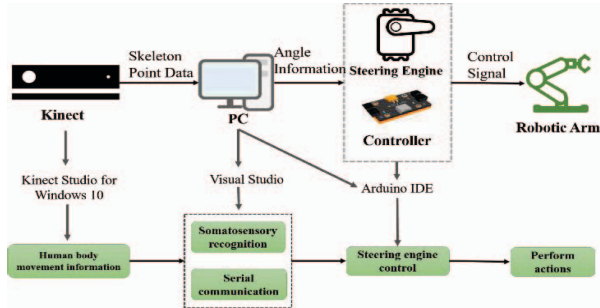


Figure 2: Somatosensory robotic arm control architecture

2.4 Control Platform

Allowing users to control the digital model of the robotic arm can simplify the operation of the system and make the system interface more intuitive[3-4]. The management and control platform of the Digital Twins manipulator is shown in Figure 4, which integrates the virtual model of the manipulator, control of the manipulator, mapping of virtual and real world, and sensor visualization interface modules.

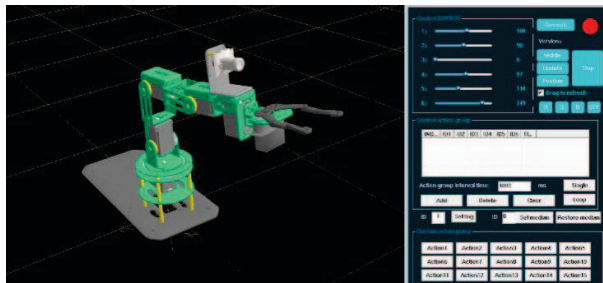


Figure 3: Robotic arm control platform

3 SYSTEM DISPLAY

Through this system, we have achieved the integration of virtual and reality, surpassing superheroes such as Iron Man, Hulk, and Captain Marvel, and lifted the "Thor's Hammer".

3.1 Overall Display

The system mainly realizes three functions:

The three-dimensional modeling software SolidWorks is used to establish a three-dimensional solid model of the robotic arm, and a reference coordinate system is assigned to each joint to realize the kinematics simulation of the robotic arm.



Figure 4: Lift Thor's Hammer

Use CAN communication to establish the simultaneous interpretation interaction between the robotic arm and the PC-side host computer, so that the robotic arm can control the current by controlling the discharge distance.

The precise control of the robotic arm is realized through fractional PID control, ensuring that the robotic arm can accurately grip objects.

Use Kinect to capture human movements and remotely transmit the information to the robotic arm platform when the person is not in contact with the robotic arm, so as to realize the function of manipulating the robotic arm through human movements.

3.2 Details Display

Figure 5 shows the effect of a hammer guiding the thunder. It is worth mentioning that the system chooses Tesla coil to create lighting. In the future, we will map the thunder and Thor's Hammer to virtual space, reflect the dynamic changes of thunder and Thor's Hammer from the real world to real time in the virtual space, in turn realize the Digital Twins of thunder.

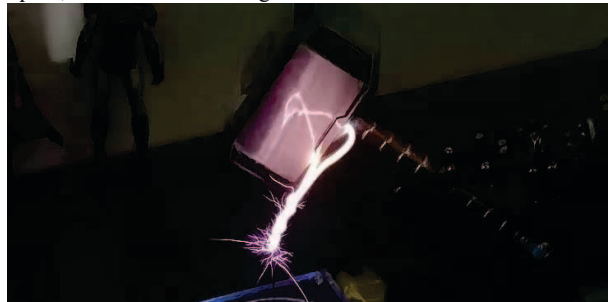


Figure 5: Thor's Hammer to guide the thunder

4 CONCLUSION AND FUTURE WORK

This system uses DT to build a digital model of the robotic arm, and uses PID, CAN and the others to control the movement and data transmission of the manipulator, and track the status of the manipulator in real time. This system realizes the perfect integration of the robotic arm and DT. With the help of the motion sensing control of the DT robotic arm, it can lift Thor's hammer accurately and safely just like Thor. We use body sense to control the robotic arm, obtain motion data through Kinect, and reflect it into the robotic arm. This system is not only enjoyable, it also simulates the process of wireless power transmission through the Tesla coil in the future, and carries out a micro-simulation of the future wireless power transmission scene. In the future, we will map lightning and Thor's hammer to virtual space, and reflect the dynamic changes of thunder and lightning in the real world, which can achieve a DT of Thor's Arm. That would be a good Thor experience.

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